

Global Water Scarcity & Freshwater Quality Analysis

Using Data Mining and Machine Learning Techniques *Using Data Mining and Machine Learning Techniques*

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1. Abstract

Freshwater scarcity and water quality are two of the most pressing global environmental challenges. This project applies data mining and machine learning techniques to two complementary questions: (1) can physicochemical measurements predict whether a water sample is safe to drink, and (2) which countries face the most severe freshwater stress. A Random Forest classifier is trained on a water-quality dataset to predict potability, while K-Means clustering groups countries by a World Bank water-stress indicator into Low, Moderate, and High stress categories. The pipeline covers data cleaning, exploratory analysis, feature scaling, supervised classification, unsupervised clustering, and interpretation of results.

2. Problem Statement & Motivation

Only a small fraction of the planet's water is both fresh and accessible, and that fraction is under increasing pressure from population growth, industrial use, and climate variability. Two distinct but related problems drive this project:

- Water safety: not all available freshwater is safe to drink — contamination from chemical and biological sources makes automated potability screening valuable for early warning systems.
- Water stress: some countries withdraw a far greater share of their renewable freshwater than others, putting them at elevated risk of future shortages.

This project treats both as data mining problems: potability prediction as a supervised classification task, and country-level stress severity as an unsupervised clustering task.

3. Data Sources

3.1 Water Quality Dataset

Source: Kaggle. Contains 9 physicochemical parameters per water sample — pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Total Organic Carbon, Trihalomethanes, and Turbidity — with a binary target variable, Potability (0 = not safe, 1 = safe).

3.2 Global Water Stress Dataset

Source: World Bank Open Data. Indicator: freshwater withdrawal as a percentage of renewable internal freshwater resources, filtered to the most recent available year (2022) for this analysis.

3.3 Metadata

Official World Bank indicator metadata, used to attach a plain-language definition of the water-stress indicator to the report and notebook.

4. Methodology

4.1 Data Cleaning & Preprocessing

- Missing numeric values in the water-quality dataset were imputed using column-wise means.
- The water-stress dataset was filtered to the 'Country Name' and 2022 columns, renamed for clarity, and rows with missing stress values were dropped.

4.2 Exploratory Data Analysis

- A correlation heatmap was generated across all water-quality features to surface multicollinearity and relationships with potability.
- A boxplot of global water-stress percentages was used to inspect the distribution and identify outlier countries before clustering.

4.3 Feature Scaling

All water-quality features were scaled to a common range using Min-Max normalization, which is important for distance- and split-based models and keeps feature magnitudes comparable.

4.4 Classification — Random Forest

- An 80/20 stratified train-test split preserved the potability class ratio in both sets.
- A Random Forest with 200 trees was trained with `class_weight='balanced'` to counteract the natural imbalance between safe and unsafe samples.
- Performance was evaluated with accuracy, precision, recall, F1-score, and a confusion matrix, plus a feature-importance ranking to identify which chemical parameters matter most.

4.5 Clustering — K-Means

- The elbow method ($k = 2$ to 5) was used to select the number of clusters based on inertia.
- K-Means with $k=3$ grouped countries by water-stress percentage into three severity bands.
- A silhouette score validated the separation quality of the resulting clusters.
- Cluster labels (Low / Moderate / High) were assigned by ranking cluster centroids by stress percentage, rather than by raw cluster index — this avoids mislabeling clusters, since K-Means cluster numbers are arbitrary and not inherently ordered by severity.

5. Results

The following metrics and artifacts summarize model performance. Exact values will populate after `code.py` is run in your environment (see the `outputs/` folder for generated charts):

Metric	Description	Where to find it
Accuracy	Overall correctness of potability predictions	Console output / <code>classification_report</code>
Precision / Recall / F1	Per-class performance, important given class imbalance	<code>classification_report</code>
Confusion Matrix	True vs. predicted potability counts	<code>outputs/confusion_matrix.png</code>
Feature Importance	Which chemical parameters drive predictions most	<code>outputs/feature_importance.png</code>
Silhouette Score	Cluster separation quality for water-stress groups	Console output

Key qualitative findings:

- Certain chemical parameters (per the feature-importance chart) have a disproportionate influence on whether a water sample is classified as safe.
- A subset of countries show extreme water-stress percentages — in some cases withdrawing more freshwater annually than is renewably replenished — placing them firmly in the High Water Stress cluster.
- Clustering provides a simple, interpretable way to triage countries by urgency for water-policy attention, rather than relying on a single raw percentage figure.

6. Limitations

- The water-quality and water-stress datasets are not linked at the country level, so potability findings and stress findings should be read as two separate analyses rather than a single integrated model.
- Clustering was performed on a single feature (stress percentage); incorporating additional indicators (population, GDP, agricultural water use) could produce richer country groupings.
- No cross-validation was used for the classifier; a single train/test split, while stratified, gives a less stable estimate of generalization than k-fold CV.
- Mean imputation for missing water-quality values is simple but can understate variance; more advanced imputation (e.g. KNN or iterative imputation) could improve robustness.

7. Corrections Made to the Original Implementation

The following issues were identified in the original code.py and corrected in this update:

Issue	Why it mattered	Fix applied
Hard-coded absolute file paths (C:/Users/LAPTOP INSIDE/...)	Script would only run on the original author's machine; breaks for graders, recruiters, or Colab users	Replaced with relative paths resolved from the script's own location
outputs/ folder assumed to exist	plt.savefig() throws FileNotFoundError on a fresh clone	Added os.makedirs('outputs', exist_ok=True) at startup
Arbitrary K-Means cluster-to-label mapping	Cluster numbers 0/1/2 are not ordered by severity; the fixed mapping {0:Low,1:Moderate,2:High} could silently mislabel every country	Clusters are now ranked by centroid value before labels are assigned, guaranteeing 'High' always means highest withdrawal %
fillna(df.mean(), inplace=True) chain	Raises a FutureWarning on current pandas versions (silent downcasting)	Reassigned result of fillna() instead of chained inplace call
No requirements.txt	Reviewers/graders can't reproduce the environment	Added requirements.txt with pinned core libraries
File names contain spaces ('Water scarcity.csv', 'Project Report.pdf')	Breaks some shell commands / CI pipelines and looks unpolished in a portfolio repo	Recommended rename to underscore_case (kept as-is in code for compatibility; noted as a follow-up)

8. Conclusion

This project demonstrates a complete, reproducible data mining pipeline spanning supervised classification and unsupervised clustering on real-world environmental data. The Random Forest model provides an interpretable, class-imbalance-aware approach to potability prediction, while the corrected K-Means clustering reliably ranks countries by water-stress severity. Together, they offer a template for combining water-quality and water-scarcity analysis for environmental and policy-relevant insight.

9. References

- Kaggle: Water Potability Dataset.
- World Bank Open Data: Freshwater withdrawal as % of renewable resources indicator, 2022.

- [scikit-learn documentation](#): RandomForestClassifier, KMeans, MinMaxScaler.